Flyback Based PFC Converter

Introduction

In the realm of electric vehicle (EV) charging systems, maintaining a high power factor is crucial for efficient and reliable operation. Previous discussions have explored various Power Factor Correction (PFC) converters, such as boost converter-based PFCs and totem pole PFCs. These designs typically employ a diode bridge rectifier at the front end, followed by a DC-DC converter, or utilize active switches to directly shape the input current. The primary goal across all these topologies is to ensure that the current drawn from the AC source is sinusoidal and in phase with the voltage, thereby achieving a unity power factor. This is often accomplished by controlling the average current or operating the converter in Discontinuous Conduction Mode (DCM), which simplifies control by eliminating the need for complex multipliers.

While boost and totem pole converters are effective, there are alternative methods to achieve regulated output voltage and unity power factor. One such method, particularly suitable for lower power applications, is the flyback-based PFC converter. This lecture delves into the architecture, operating principles, and advantages of using a flyback converter for PFC. Unlike boost converters, flyback converters inherently offer galvanic isolation between the input and output, a significant benefit in many power supply applications. Understanding its operation, especially in DCM, reveals how it can simplify control and effectively draw a sinusoidal current from the source, making it a valuable topology in the charging infrastructure landscape.

Core Concepts

Term	Definition	Significance
`PFC (Power Factor Correction)`	A technique used to improve the power factor of an AC power supply, making the current waveform more sinusoidal and in phase with the voltage.	Ensures efficient power delivery, reduces harmonic distortion, and minimizes losses in the power grid.
`Flyback Converter`	A type of DC-DC converter that uses a transformer to store energy during the	Provides galvanic isolation between input and output, making it suitable for

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	switch-on period and transfer it to the output during the switch-off period.	applications requiring safety separation.
`DCM (Discontinuous Conduction Mode)`	An operating mode where the inductor current (or transformer primary current) falls to zero before the end of each switching cycle.	Simplifies control by inherently shaping the input current to follow the input voltage, often eliminating the need for current multipliers.
`Volt-Second Balance`	A principle stating that the average voltage across an inductor over a complete switching cycle must be zero in steady-state operation.	Essential for determining the steady-state duty ratio and ensuring stable operation of DC-DC converters.
`Duty Ratio (d(t))`	The fraction of the switching period during which the main switch of a converter is turned on. It can vary with time (d(t)).	Controls the energy transfer and output voltage regulation in converters; its variation is key for PFC.
`L_critical (Critical Inductance)`	The maximum inductance value for which a converter will operate in Discontinuous Conduction Mode (DCM) under specific load and input conditions.	Crucial for design, as selecting an inductance below this value guarantees DCM operation, simplifying PFC control.
`Resistive Input Impedance`	A characteristic where a converter appears as a pure resistive load to the AC source, meaning the input current is proportional to the input voltage.	Achieves unity power factor, as the current drawn is sinusoidal and in phase with the source voltage.
`Isolation`	The electrical separation between the input and output	Enhances safety by preventing direct current paths between the source

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	circuits, typically achieved using a transformer.	and the load, protecting users from electrical shock.
`Voltage Follower Mode`	A control strategy where the converter's input current naturally follows the shape of the input voltage, often achieved through DCM operation.	Simplifies the control loop for PFC, as the current shaping is an inherent characteristic of the operating mode.

Detailed Analysis

1. Flyback Converter Architecture

The flyback-based PFC converter typically begins with a diode bridge rectifier, which converts the incoming AC voltage into a rectified DC voltage. This rectified voltage, denoted as `Vd`, serves as the input to the flyback converter stage. The flyback converter itself consists of a main switch (S1), a coupled inductor (acting as a transformer with primary and secondary windings), a rectifier diode (DB) on the secondary side, an output capacitor (C), and the load (RL).

The coupled inductor is central to the flyback's operation. It features a primary winding (P) and a secondary winding (S), with a turns ratio of `n:1` (where `n` is the turns ratio of primary to secondary). Dot polarities are crucial: when the primary side dot is positive, the secondary side dot is also positive. However, in a flyback configuration, the secondary winding is connected such that its dot is connected to the output capacitor and load via the diode DB, which is reverse-biased when the primary switch S1 is on. The output voltage `Vo` is maintained across the capacitor and load. The input AC voltage `Vs` is a sinusoidal waveform, `Vs,pk * sin(ωt)`, and the rectified voltage `Vd` is `|Vs|`.

2. Operating Principle: Switching States

The flyback converter operates in two main states within each high-frequency switching period (`Ts`), determined by the state of the main switch S1. The switching frequency (`fsw`) is significantly higher than the line frequency (`fs`, typically 50 or 60 Hz).

- **S1 ON Period (d(t)Ts):**
- When switch S1 is turned on, the rectified input voltage `Vd` is directly applied across the primary winding of the coupled inductor.

- Due to the dot polarity arrangement, a voltage is induced across the secondary winding. However, the diode DB is reverse-biased during this period, preventing current flow to the output.
- Energy is stored in the primary winding's inductance `L`. The current in the primary winding rises linearly with a slope proportional to `Vd/L`.
- During this phase, the output capacitor `C` supplies the load `RL`, as no energy is transferred from the input to the output.
- **S1 OFF Period ((1-d)Ts):**
- When switch S1 is turned off, the primary current path is interrupted. The energy stored in the coupled inductor's core must be released.
- The voltage polarity across the windings reverses. The secondary side dot becomes negative, and the non-dot terminal becomes positive, forward-biasing the diode DB.
- Current flows from the secondary winding through the diode DB to charge the output capacitor `C` and supply the load `RL`.
- The current in the secondary winding decreases linearly, with a slope proportional to `-Vo/L_secondary` (or `-n*Vo/L_primary` when referred to the primary side).
- The core de-energizes, and the flux built up during the S1 ON period decays.

3. Volt-Second Balance and Duty Ratio

To maintain a constant output DC voltage `Vo`, the duty ratio `d(t)` of switch S1 must be controlled. This control is derived using the principle of volt-second balance across the primary inductor over a switching period. Since the input `Vd` is a rectified AC voltage, `d(t)` must vary over the line cycle.

Applying volt-second balance, the average voltage across the primary inductor over a switching period `Ts` must be zero.
During `d(t)Ts`, the voltage across the primary is `Vd`.
During `(1-d)Ts`, the voltage across the primary, referred from the secondary, is `-n * Vo`.
Therefore, `Vd * d(t)Ts + (-n * Vo) * (1-d(t))Ts = 0`.
Simplifying this equation, we get:
`Vd * d(t) = n * Vo * (1 - d(t))`
`Vd * d(t) = n * Vo - n * Vo * d(t)`
`d(t) * (Vd + n * Vo) = n * Vo`
`d(t) = (n * Vo) / (Vd + n * Vo)`
This equation shows how the duty ratio `d(t)` must vary with the instantaneous rectified input voltage `Vd` to maintain a constant output voltage `Vo`.

4. Discontinuous Conduction Mode (DCM) Operation

For PFC applications, operating the flyback converter in Discontinuous Conduction Mode (DCM) offers significant advantages. In DCM, the inductor current (primary current) falls to zero before the end of the switching period. This means there is a third state where both S1 and DB are off, and no current flows through the inductor.

- **Inductor Current Waveform in DCM:**
- During `d(t)Ts` (S1 ON), the primary current rises linearly from zero to a peak value `ip`. The slope is `Vd/L`.
- During `d2(t)Ts` (DB ON), the secondary current (referred to primary) falls linearly from `n*ip` to zero. The slope is `-n*Vo/L`.
- During the remaining `(1 - d(t) - d2(t))Ts` period, the current is zero.
- **Peak Inductor Current (`ip`) and PFC:**
  The peak inductor current `ip` can be expressed as:
  `ip = (Vd * d(t) * Ts) / L`
  Substituting the derived `d(t)` into this equation, it can be shown that `ip` varies proportionally with `Vd`. This is a crucial characteristic for PFC. If the peak current drawn from the input follows the rectified input voltage, then the average current drawn over a switching cycle will also follow the rectified input voltage.
- **Resistive Input Impedance:**
  In DCM, the average input current over a switching period (`iL_avg`) can be derived. It is proportional to `Vd`. This implies that the converter presents a purely resistive load to the input source.
  `iL_avg = (1/Ts) * (1/2) * ip * (d(t)Ts + d2(t)Ts)`
  After detailed derivation, it can be shown that `iL_avg` is proportional to `Vd / R_effective`, where `R_effective` is an equivalent input resistance.
  This `R_effective` is a function of `L`, `d(t)`, and `Ts`.
  The fact that `iL_avg` follows `Vd` means that the input current from the AC source will be sinusoidal and in phase with the input voltage, achieving unity power factor.
- **Advantages of DCM for PFC:**
- **Simpler Control:** DCM operation inherently provides voltage follower characteristics, meaning the input current naturally tracks the input voltage. This eliminates the need for a complex current multiplier in the control loop, simplifying the controller design and reducing computational requirements.
- **No Reverse Recovery Issues:** The output diode turns off when its current naturally reaches zero, avoiding reverse recovery losses and stress.

5. Critical Inductance (L_critical)

To ensure DCM operation, the inductance `L` of the primary winding must be carefully selected. There is a critical inductance value, `L_critical`, below which the converter will always operate in DCM. If `L` is greater than `L_critical`, the converter will enter Continuous Conduction Mode (CCM), where the inductor current never falls to zero.

The `L_critical` value is derived by setting the total conduction time (`d(t)Ts + d2(t)Ts`) equal to the switching period `Ts`. The lecture provides a formula for `L_critical` (referred to the primary side):
`L_critical = (n^2 * RL * Ts) / (4 * (1 + (n * Vo / Vs,peak))^2)`
This formula indicates that `L_critical` depends on the turns ratio `n`, load resistance `RL`, switching period `Ts`, output voltage `Vo`, and peak input voltage `Vs,peak`.

For guaranteed DCM operation, the chosen inductance `L` must be less than `L_critical`. To ensure DCM under all operating conditions, `L` should be less than the minimum `L_critical` value. The minimum `L_critical` occurs under specific conditions:
* **Minimum Load Resistance (RL_min):** Corresponds to the peak power demand from the converter.
* **Minimum Input Peak Voltage (Vs,peak_min):** The lowest expected AC input voltage.
Therefore, `L` should be selected such that `L < L_critical_min` (calculated at `RL_min` and `Vs,peak_min`).

6. Advantages and Limitations

- **Isolation:** A key advantage of the flyback converter is the inherent galvanic isolation provided by the transformer. This is crucial for safety in many applications, including EV charging, where separation between the grid and the vehicle's battery is desired.
- **Voltage Follower Mode:** As discussed, DCM operation allows the flyback converter to naturally operate in a voltage follower mode, simplifying PFC control.
- **Low Power Applications:** Flyback converters are generally well-suited for low to medium power applications. For high-power applications, the size of the coupled inductor (transformer) can become prohibitively large, making other topologies like boost or totem pole PFCs more practical.
- **Buck-Boost Derivative:** The flyback converter is essentially a buck-boost converter with an isolated output. A non-isolated buck-boost converter can

also be used for PFC by operating it in DCM, achieving a resistive input impedance and unity power factor.

Key Takeaways

- [ ] Flyback PFC converters use a diode bridge rectifier followed by an isolated flyback stage to achieve power factor correction.
- [ ] The primary goal is to draw a sinusoidal current from the AC source, in phase with the voltage, for unity power factor.
- [ ] Operating the flyback converter in Discontinuous Conduction Mode (DCM) simplifies PFC control by inherently making the input current follow the input voltage.
- [ ] In DCM, the inductor current falls to zero within each switching cycle, creating a "voltage follower" characteristic.
- [ ] Volt-second balance is used to determine the time-varying duty ratio `d(t)` required to maintain a constant output voltage.
- [ ] The peak inductor current in DCM is proportional to the instantaneous rectified input voltage, leading to a resistive input impedance.
- [ ] The inductance `L` must be chosen to be less than a critical inductance `L_critical` to ensure DCM operation under all load and input voltage conditions.
- [ ] Flyback PFC converters offer galvanic isolation between input and output, enhancing safety.
- [ ] They are generally preferred for low to medium power applications due to the size constraints of the coupled inductor at higher power levels.
- [ ] Buck-boost converters, being the non-isolated counterpart, can also achieve PFC when operated in DCM.

Conclusion

The flyback-based PFC converter, particularly when operated in Discontinuous Conduction Mode, offers a compelling solution for power factor correction in low to medium power applications. Its inherent galvanic isolation provides a significant safety advantage, making it suitable for sensitive applications like EV charging where separation between the grid and the vehicle's high-voltage battery is paramount. The simplified control strategy afforded by DCM, which eliminates the need for complex current multipliers, reduces the computational burden on controllers and can lead to more cost-effective designs.

While the flyback topology excels in these areas, its scalability to very high power levels is limited by the physical size of the coupled inductor. This understanding

helps engineers select the appropriate PFC topology based on power requirements, isolation needs, and control complexity. As EV charging infrastructure continues to evolve, the principles learned from single-phase PFC converters, including the flyback, form a foundational understanding for tackling more complex three-phase systems, which will be the focus of future discussions. The ability to efficiently draw power from the grid while maintaining high power quality remains a cornerstone of modern power electronics design.